Project Summary:

"Customer Support Ticket Analysis System"

✓ Purpose:

The purpose of this project was to simulate a real-world customer support environment and understand how raw ticket data can be transformed into meaningful insights. The goal was to build a system that can store, clean, and analyze customer issues to identify patterns in service quality and customer feedback.

✓ Tools & Technologies Used:

- Colab & Python
    - Core Concepts:
        - Lists, Dictionaries, Sets
        - Functions
        - Loops & Conditional Statements
        - String Manipulation & Text Processing

✓ Result:

- Successfully developed a Python-based system that:
- Cleaned and standardized unstructured ticket data for consistency
- Enabled dynamic addition of tickets with proper validation and auto-incremented IDs
- Identified key patterns using keyword-based analysis (e.g., "poor", "slow", "good", "excellent")
- Provided priority distribution insights (High, Medium, Low) for better workload understanding
- Highlighted complex issues by detecting the longest ticket descriptions
- Extracted unique words to understand commonly used customer language

✓ Challenges Faced:

- Handling inconsistent and messy text data (extra spaces, punctuation, mixed cases)
- Ensuring data standardization for accurate analysis
- Implementing input validation for user-added tickets
- Designing logic to analyze text without advanced libraries
- Maintaining a structured format using dictionary of lists

✓ Conclusion:

This project demonstrates how structured data processing and basic text analysis can convert raw customer feedback into actionable insights. It highlights the importance of data cleaning, categorization, and keyword analysis in understanding customer behavior. Overall, the system provides a simple yet effective foundation for real-world applications such as customer feedback analysis, support optimization, and service improvement strategies.